

Installing MLDB on different platforms

Mldb provides a Web based interface for the easiest implementation and hands-on experience. A free session of MLDB can be experienced for 90 minutes using a Web based panel after signing up (registration) on <https://mldb.ai/#signup>. There are many demos and a lot of documentation available so that a cloud based MLDB can be worked out without installation on the local system. Even the self-created datasets can be uploaded on this hosted session.

Editions for local installation

There are two editions of MLDB that are free, and are distributed as community and enterprise editions. To run the MLDB enterprise edition, you need to enter the licence key to activate the software. A licence key can be created for first-time users on signing up at https://mldb.ai/#license_management and filling the required details in the registration form.

	Community edition	Enterprise edition (Free trial)
MLDB	Available	Available
Licensing	Apache License v2.0	Non-commercial
Cost / Pricing	Free	Free
Issues and support	GitHub issues	MLDB support
Download Instructions	https://github.com/mldbai/mldb/blob/master/Building.md	https://docs.mldb.ai/doc/builtin/Running.md.html#packages

Docker image

The classical MLDB distribution is the Docker image. While other distributions are available for virtual machines, the Docker image is executed as a container. This method is used for Linux flavours or private cloud deployments.

After the installation of Docker, the MLDB container is launched with a pre-specified mapped directory.

```
$ mkdir </my/system/path/myMLDBdata>
```

With the execution of the following commands, a port can be set using the `mldbport` parameter.

```
docker run --rm=true \
-v </my/system/path/myMLDBdata>:/mldb_data \
-e MLDB_IDS="id" \
-p 127.0.0.1 (IP-Address):<mldbport>:80 (Port) \
quay.io/mldb/mldb:latest
```

For security and overall integrity, a tunnel is established for remote servers as in the following instruction using the

SSH tunnel:

```
$ ssh -f -o ExitOnForwardFailure=yes <user>@<remotehost> -L <localport>:127.0.0.1 (IP-Address):<mldbport> -N
```

Once the message 'MLDB Ready' is viewed, the browsing and activation of MLDB can be done on a Web browser using the URL `http://localhost:<localport>`.

Virtual appliance

The installation of MLDB for virtualisation is very easy. The virtual application (OVA file) is available so that it can be imported using VirtualBox or any other virtualisation software.

After downloading VirtualBox from <https://www.virtualbox.org/wiki/Downloads>, the OVA file of MLDB Appliance at <http://public.mldb.ai/mldb.ova> can be imported.

Simply double-click the OVA file or select 'Import Appliance' in the File Menu of VirtualBox and finally point out to the downloaded MLDB OVA distribution.

The default user name to log in on OVA is 'ubuntu' and the password for successful authentication is 'mldb'.

After these steps, the MLDB instance can be executed using the URL `http://localhost:8080/` on any Web browser.

Cloud servers

MLDB can be installed and deployed on the cloud environment of Amazon Web Services (AWS). The Amazon Machine Image (AMI) is available for deployment on AWS, and can be easily attached in the dashboard of Amazon Cloud.

The steps needed for attaching and configuring AMI in AWS are as follows:

1. Create an Amazon Web Services (AWS) account on <http://aws.amazon.com/>
2. 'Create Instance Wizard' should be selected from 'N', the Virginia Zone on the AWS dashboard.
3. Select Amazon Machine Image (AMI).
4. From Community AMIs, search 'Datacratic MLDB'.
5. Launch the latest AMI for MLDB.
6. Any instance can be selected from AWS depending upon the usage and load of the application.
7. Configure AWS instance details.
8. Add up the storage or memory parameters in Amazon.
9. Name the remote machine.
10. Integrate SSH Port 22 with the machine.
11. Launch the instance.
12. Establish a security tunnel.
13. Activate MLDB. 

By: Dr Gaurav Kumar

The author is the MD of Magma Research and Consultancy Pvt Ltd, Ambala. He delivers expert lectures and conducts workshops on the latest technologies and tools. He can be contacted at kumargaurav.in@gmail.com. His personal website is at www.gauravkumarindia.com.